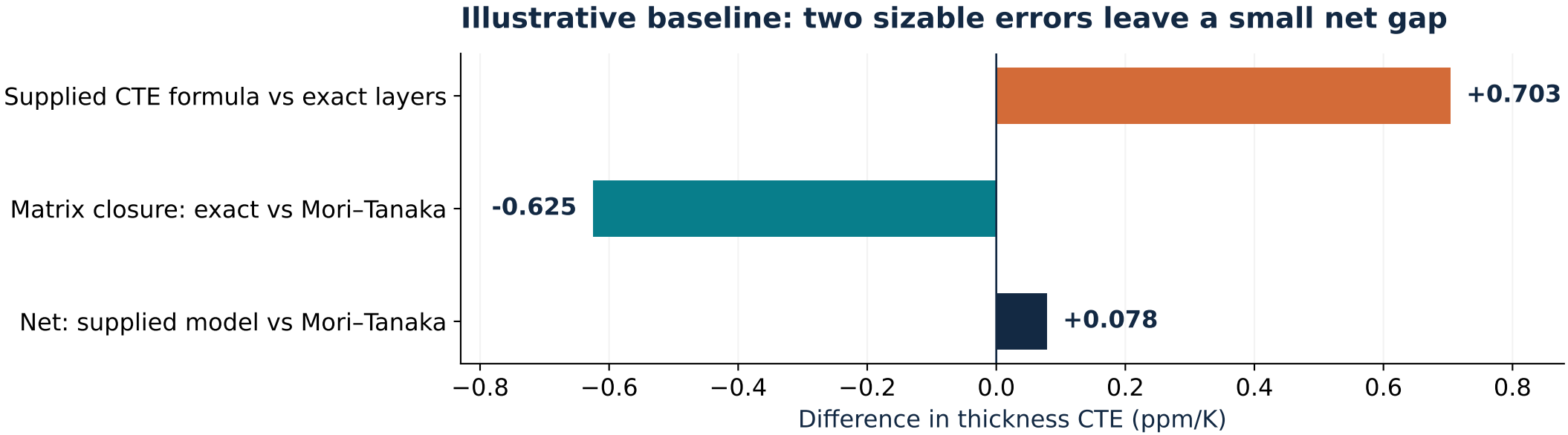
Model errors can cancel and hide disagreement

Useful review priority: validate the filled-matrix model before refining layer averaging



43.4%

of 8,192 illustrative cases had |net gap| ≤0.2 ppm/K

while each opposing component was ≥0.5 ppm/K in magnitude.

MODEL CHOICE IMPACT

Thickness modulus: 13.9% median matrix-closure gap; in-plane modulus: 7.0%.

Layer-averaging gap in interlaminar shear: 0.23% median.

Scenario ranges are illustrative, not calibrated probabilities or measured material behavior.

The 0.2 and 0.5 ppm/K thresholds are exploratory screens; model agreement is not experimental validation.